

High-Priority Sales Order Communication & Orchestration Agent

Developer Design Specification | Demo Scope

Purpose: Build a simple demo agent that manages the approval communication Sales → Planning → Production Scheduler → Planning → Sales for a new high-priority sales order. The agent coordinates requests, notifications, responses, status, and audit history. No Company Planner or external planning system is required for this demo.

1. Demo Objective and Scope

- Demonstrate a human-in-the-loop workflow across Sales, Planning, and Production Scheduler teams.
- Use connected sample data across Customer → Sales Order → FG → SFG → RAW1/RAW2 → production/resource situation.
- Agent must not autonomously confirm the sales order or reschedule production. Team decisions remain explicit Confirm / Reject actions.
- For the demo, stock, order, BOM, and capacity data can come from mock JSON, an in-memory database, or simple demo tables.

2. Connected Sample Data

Entity	ID	Description	Qty / Stock	Priority	Status
Customer	CUST-1001	ABC Automotive			Active
High-priority SO	SO-005	FG-100 Finished Product	100	HIGH	Awaiting feasibility
Normal SO	SO-004	FG-100 Finished Product	100	NORMAL	Planned / candidate to move
FG	FG-100	Finished Product	Stock 0		Shortage
SFG	SFG-200	Semi-Finished Product	Stock 0		Shortage
RAW1	RAW-1	Primary Raw Material	Stock 150 / Need 105		Available
RAW2	RAW-2	Secondary Raw Material	Stock 20 / Need 5		Available

Demo dependency: CUST-1001 → SO-005 → FG-100 → SFG-200 → RAW-1 + RAW-2. Because FG and SFG are unavailable but both raw materials are available, Planning asks Production Scheduler whether SO-004 can be rescheduled so SO-005 can be produced first.

3. Roles and Responsibilities

Role	Responsibility in Demo	Decision
Sales / Customer Service	Create feasibility request for SO-005; wait for Planning response; confirm SO only after Planning confirms.	Initiate; final SO confirmation
Planning	Check FG/SFG/raw stock; if feasible from material perspective, request Production Scheduler decision.	Confirm / Reject back to Sales
Production Scheduler	Check existing orders and resource schedule; evaluate moving SO-004 behind SO-005.	Confirm / Reject rescheduling possibility
Orchestration Agent	Route work, send notifications, preserve context, update status, prevent invalid transitions, maintain audit trail.	No independent business approval

4. End-to-End Workflow

Step 1 - Sales → Planning

Sales opens SO-005 and submits “Check High-Priority Confirmation Possibility”. Agent creates request REQ-001 and notifies Planning.

System outcome: Planning receives Customer, SO, FG, quantity, requested date, priority, and request reason. Actions: Review / Confirm / Reject.

Step 2 - Planning stock check

Planning reviews FG-100, SFG-200, RAW-1 and RAW-2. Demo result: FG=0, SFG=0, RAW1 and RAW2 available.

System outcome: Planning does not reject because production is still materially possible. Planning selects “Request Production Rescheduling Check”.

Step 3 - Planning → Production Scheduler

Agent creates child request SCH-001 linked to REQ-001 and notifies Production Scheduler.

System outcome: Message shows high-priority SO-005 and affected normal order SO-004. Question: “Can SO-004 be rescheduled and SO-005 inserted first?” Actions: Confirm / Reject.

Step 4 - Production Scheduler decision

Scheduler reviews demo resource load and order sequence. For happy-path demo, Scheduler confirms the change is possible.

System outcome: Scheduler submits Confirm plus proposed production date and optional comment. Agent returns the decision to Planning.

Step 5 - Planning → Sales

Planning receives Scheduler confirmation, reviews the complete chain, and selects Confirm.

System outcome: Agent notifies Sales that SO-005 can be confirmed, including the confirmed/proposed date and decision trail.

Step 6 - Sales confirmation

Sales reviews Planning confirmation and selects “Confirm Sales Order”.

System outcome: REQ-001 status becomes COMPLETED. Agent stores final outcome and audit log.

5. Developer Workflow / State Flow

SALES

| Submit feasibility request



AGENT: REQ-001 = WAITING_FOR_PLANNING

| Notify Planning



PLANNING

| Check FG / SFG / RAW1 / RAW2

| Request production rescheduling check



AGENT: SCH-001 = WAITING_FOR_PRODUCTION

| Notify Production Scheduler



PRODUCTION SCHEDULER

| Confirm / Reject rescheduling SO-004 for SO-005



AGENT: return decision to Planning



PLANNING

| Confirm / Reject feasibility

▼
AGENT: return decision to Sales
▼
SALES
| Confirm SO-005 (only when Planning = CONFIRMED)
▼
AGENT: REQ-001 = COMPLETED

6. Required Status Model and Business Rules

Status	Owner / Waiting For	Allowed Next State
DRAFT	Sales	WAITING_FOR_PLANNING
WAITING_FOR_PLANNING	Planning	WAITING_FOR_PRODUCTION / REJECTED_BY_PLANNING
WAITING_FOR_PRODUCTION	Production Scheduler	PRODUCTION_CONFIRMED / REJECTED_BY_PRODUCTION
PRODUCTION_CONFIRMED	Planning	PLANNING_CONFIRMED / REJECTED_BY_PLANNING
PLANNING_CONFIRMED	Sales	COMPLETED
REJECTED_BY_PRODUCTION	Planning	REJECTED_BY_PLANNING or resubmit scheduler request
REJECTED_BY_PLANNING	Sales	Closed / new request
COMPLETED	None	Final

- Sales Order confirmation action must be disabled unless REQ-001 is PLANNING_CONFIRMED.
- Planning Confirm must be disabled when a scheduler check is required but Production Scheduler has not confirmed.
- Every Confirm or Reject action must capture acting role/user, timestamp, comment, and previous/new status.
- Reject requires a reason. Confirm can allow an optional comment.
- A child scheduling request must always reference its parent feasibility request.

7. Minimal Data Objects / API Payloads

7.1 Feasibility request

```
{
  "requestId": "REQ-001",
  "customerId": "CUST-1001",
  "salesOrder": "SO-005",
  "fg": "FG-100",
  "quantity": 100,
  "priority": "HIGH",
  "requestedDeliveryDate": "<demo-date>",
  "status": "WAITING_FOR_PLANNING",
  "requestedByRole": "SALES"
}
```

7.2 Planning stock result

```
{
  "requestId": "REQ-001",
  "fg": {"material": "FG-100", "required": 100, "available": 0},
  "sfg": {"material": "SFG-200", "required": 100, "available": 0},
  "rawMaterials": [
    {"material": "RAW-1", "required": 105, "available": 150, "status": "AVAILABLE"},
    {"material": "RAW-2", "required": 5, "available": 20, "status": "AVAILABLE"}
  ],
  "planningAction": "REQUEST_PRODUCTION_CHECK"
}
```

7.3 Scheduler request / decision

```
{
  "scheduleRequestId": "SCH-001",
```

```

"parentRequestId": "REQ-001",
"highPrioritySalesOrder": "SO-005",
"orderToReschedule": "SO-004",
"question": "Can SO-004 be rescheduled so SO-005 can be produced first?",
"decision": "CONFIRM",
"proposedProductionDate": "<demo-date>",
"comment": "Rescheduling is possible for the demo scenario."
}

```

7.4 Final Planning response

```

{
  "requestId": "REQ-001",
  "decision": "CONFIRM",
  "confirmedDate": "<demo-date>",
  "comment": "Material situation checked and Production Scheduler confirmed rescheduling.",
  "status": "PLANNING_CONFIRMED"
}

```

8. Notification Requirements

Trigger	Recipient	Notification Content	Actions
Sales submits REQ-001	Planning	High-priority feasibility request with SO/customer/product/date/qty.	Confirm / Reject / Request Production Check
Planning requests schedule check	Production Scheduler	SO-005 request + SO-004 candidate + material check summary.	Confirm / Reject
Scheduler responds	Planning	Scheduler decision, proposed date, comment.	Confirm to Sales / Reject
Planning responds	Sales	Final feasibility decision and confirmed/proposed date.	Confirm Sales Order
Sales confirms	Planning + Production Scheduler	Request completed and SO-005 confirmed.	View only

9. Suggested Demo Screens

- Sales Worklist: customer, SO, FG, quantity, priority, requested date, request status, “Check Possibility” button.
- Planning Inbox: request details, connected material tree FG → SFG → RAW1/RAW2, stock values, decision buttons.
- Production Scheduler Inbox: high-priority order, normal order to move, simple resource/load panel, Confirm/Reject buttons.
- Request Timeline: Sales request → Planning stock check → Scheduler decision → Planning decision → Sales confirmation.
- Notification Center: unread requests and deep-link to the associated request screen.

10. Orchestration Pseudocode

```

onSalesSubmit(request):
  createRequest(request, status="WAITING_FOR_PLANNING")
  notify(PLANNING, request)

onPlanningStockCheck(requestId, stockResult):
  saveStockResult(requestId, stockResult)
  if planningRejects:
    setStatus(requestId, "REJECTED_BY_PLANNING")
    notify(SALES, decision)
  else:
    createScheduleRequest(parent=requestId, status="WAITING_FOR_PRODUCTION")
    setStatus(requestId, "WAITING_FOR_PRODUCTION")
    notify(PRODUCTION_SCHEDULER, scheduleRequest)

onSchedulerDecision(scheduleRequestId, decision):

```

```

saveDecision(scheduleRequestId, decision)
if decision == "CONFIRM": setParentStatus("PRODUCTION_CONFIRMED")
else: setParentStatus("REJECTED_BY_PRODUCTION")
notify(PLANNING, decision)

onPlanningDecision(requestId, decision):
    require schedulerDecision == "CONFIRM" when scheduler check was requested
    setStatus(requestId, "PLANNING_CONFIRMED" if CONFIRM else "REJECTED_BY_PLANNING")
    notify(SALES, decision)

onSalesConfirm(requestId):
    require status == "PLANNING_CONFIRMED"
    markSalesOrderConfirmed()
    setStatus(requestId, "COMPLETED")
    notify(PLANNING, completion)
    notify(PRODUCTION_SCHEDULER, completion)

```

11. Demo Acceptance Criteria

- Sales can submit a high-priority feasibility request and Planning receives a notification.
- Planning can view the connected FG/SFG/RAW material situation and send a scheduler request.
- Production Scheduler can Confirm or Reject the rescheduling request.
- Planning automatically receives the Scheduler response and can Confirm/Reject back to Sales.
- Sales cannot confirm SO-005 before Planning has confirmed.
- After Sales confirmation, the workflow is COMPLETED and the end-to-end audit trail is visible.
- A rejection at Planning or Production is clearly returned to the preceding team with the rejection reason.
- All screens show the same request ID/linkage so the conversation is one traceable business case.

12. Demo Boundary

This specification intentionally excludes Company Planner, ATP engines, SAP integration, automatic rescheduling, optimization, and real production transactions. The goal is to validate the agent orchestration, human approvals, connected demo data, notification flow, status control, and traceability. Real system APIs can replace the mock data services later without changing the core approval flow.